Discrete log | Pohlig-Hellman algorithm.

$\boxed{G = \langle g \rangle}$ $\log_g h$ for some $a \in G$.

Suppose the order of G is $\boxed{N = p_1^{e_1} p_2^{e_2} \cdots p_t^{e_t}}$

where $p_1, \ldots, p_t$ are distinct primes.

- Not discussed: to find $\log_g h$ takes $O(\sqrt{N})$ operations.

The discrete log problem $g^x = h$ can be solved in

$$O\left(\sum_{i=1}^{t} O\left(\sqrt{p_i^{e_i}}\right) + \log N\right) \text{ steps}$$

using the following procedure.

(1) For each $1 \le i \le t$, let

$$g_i = g^{N/p_i^{e_i}} \quad \& \quad h_i = h^{N/p_i^{e_i}}$$

Solve $\boxed{g_i^y = h_i}$ (i.e. find $\log_{g_i} h_i$)

Denote by y_i the sol^n of $g_i^y = h_i$ mod $p_i^{e_i}$.

(2) Use the Chinese remainder th^m to solve

$$x = y_i \mod p_i^{e_i} \qquad i = 1, 2, \ldots t. \quad (*)$$

$x \in \mathbb{Z}_N$ $\quad 0 \le x \le N-1$

Proof: The 1st part is clear. Suppose by CRT. $0 \le x < N = \prod p_i^{e_i}$ in $(*)$

Suppose $\boxed{x = y_i + p_i^{e_i} z_i}$ for some z_i. (using $(*)$)

$$\left(g^{x N/p_i^{e_i}}\right) = (g^x)^{N/p_i^{e_i}} = g^{(y_i + p_i^{e_i} z_i)\, N/p_i^{e_i}} = \left(g^{N/p_i^{e_i}}\right)^{y_i} \cdot g^{N z_i}$$

Since $\boxed{g^N = e}$, so $(g^x)^{N/p_i^{e_i}} = \left(g^{N/p_i^{e_i}}\right)^{y_i} = g_i^{y_i}$

$$= h_i$$

$$= h^{N/p_i^{e_i}}$$

So $\frac{N}{p_i^{e_i}} x \equiv \frac{N}{p_i^{e_i}} \log_g h \pmod{N_i}$

$\log_g h^{N/p_i^{e_i}} = x \frac{N}{p_i^{e_i}}$

i.e. $x \frac{N}{p_i^{e_i}} = \frac{N}{p_i^{e_i}} \log_g h$

Observe $\frac{N}{p_1^{e_1}}, \frac{N}{p_2^{e_2}}, \ldots, \frac{N}{p_t^{e_t}}$, the GCD is 1.

$\boxed{2\times3\times5, \quad 3\times5\times7, \quad 2\times7\times11}$

So $\exists\ c_1, c_2, \ldots, c_t \in \mathbb{Z}$ s.t.

$$\sum \frac{N}{p_i^{e_i}} \cdot c_i = 1$$

$$x = \sum \frac{N}{p_i^{e_i}} \cdot c_i \cdot x = \sum c_i \left(\frac{N}{p_i^{e_i}} \log_g(h)\right) = \log_g h \quad \text{mod } N.$$

Elliptic curve

$y^2 = p(x)$ where $p(x)$ is a cubic in x

Normal ~~Standard~~ form of $p(x) = x^3 + ax + b$.

$$\boxed{y^2 = x^3 + ax + b}$$

$$y^2 + a_1 xy + a_3 y = x^3 + a_2 x^2 + a_4 x + a_6. \qquad a_2 = 0$$

Weierstrass form

One can convert the Weistrass form into the normal form.

• Projective plane: $\mathbb{R}^3 \setminus \{(0,0,0)\}$

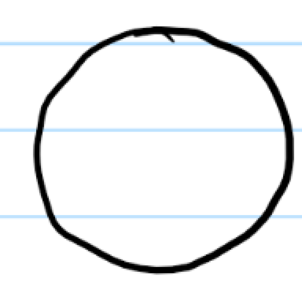
Define the relation $\sim$ by $(x,y,z) \sim (x',y',z')$

if $(x,y,z) = \lambda(x',y',z')$ for some $\lambda \neq 0$

- $\sim$ is an equivalence relation.
- Two points of $\mathbb{R}^3 \setminus \{(0,0,0)\}$ are in the same class if & only if they lie on the same straight line through the origin.

$$\mathbb{P}^2(\mathbb{R}) = \{[(x,y,z)] : (x,y,z) \in \mathbb{R}^3 \setminus \{(0,0,0)\}\}$$

$[P]$ means the equivalence class containing the point $P \in \mathbb{R}^3 \setminus \{(0,0,0)\}$.



$(x,y) \in \mathbb{R}^2$ associate it to $[(x,y,1)] \in \mathbb{P}^2(\mathbb{R})$

$$\mathbb{R}^2 \hookrightarrow \mathbb{P}^2(\mathbb{R})$$

$\mathbb{P}^1(\mathbb{R})$

$\mathbb{R} \hookrightarrow \mathbb{P}^1(\mathbb{R})$

$x \longmapsto [(x,1)]$

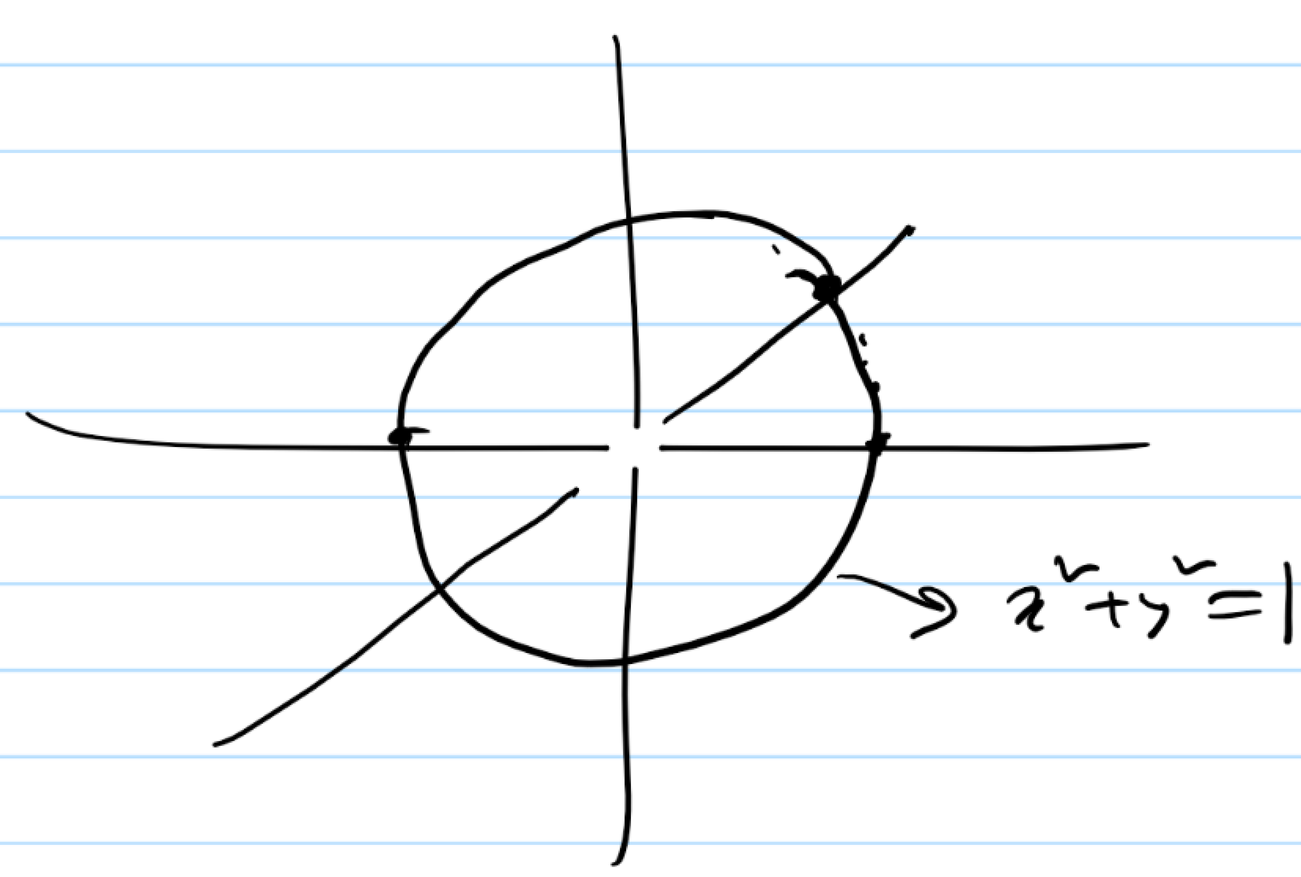
$[(1,0)] = \mathbb{P}^1(\mathbb{R}) \setminus \mathbb{R}.$

$$\mathbb{P}^2(\mathbb{R}) \ni \begin{cases} [(1,0,0)] & \text{is not in } \mathbb{R}^2 \\ [(0,1,0)] & \text{'' \quad '' } \mathbb{R}^2 \end{cases}$$

$$\mathbb{R}^1 \hookrightarrow \{[(x,y,0)] : (x,y) \neq (0,0)\} = \frac{\mathbb{R}^2 \setminus \{(0,0)\}}{\sim} = \mathbb{P}^1(\mathbb{R})$$

$x \longmapsto [(x,1,0)].$

$$\mathbb{P}^2(\mathbb{R}) = \mathbb{R}^2 \sqcup \mathbb{P}^1(\mathbb{R})$$



$$\mathbb{R}^1 \cong \{[(x,1,0)] : x \in \mathbb{R}\}$$

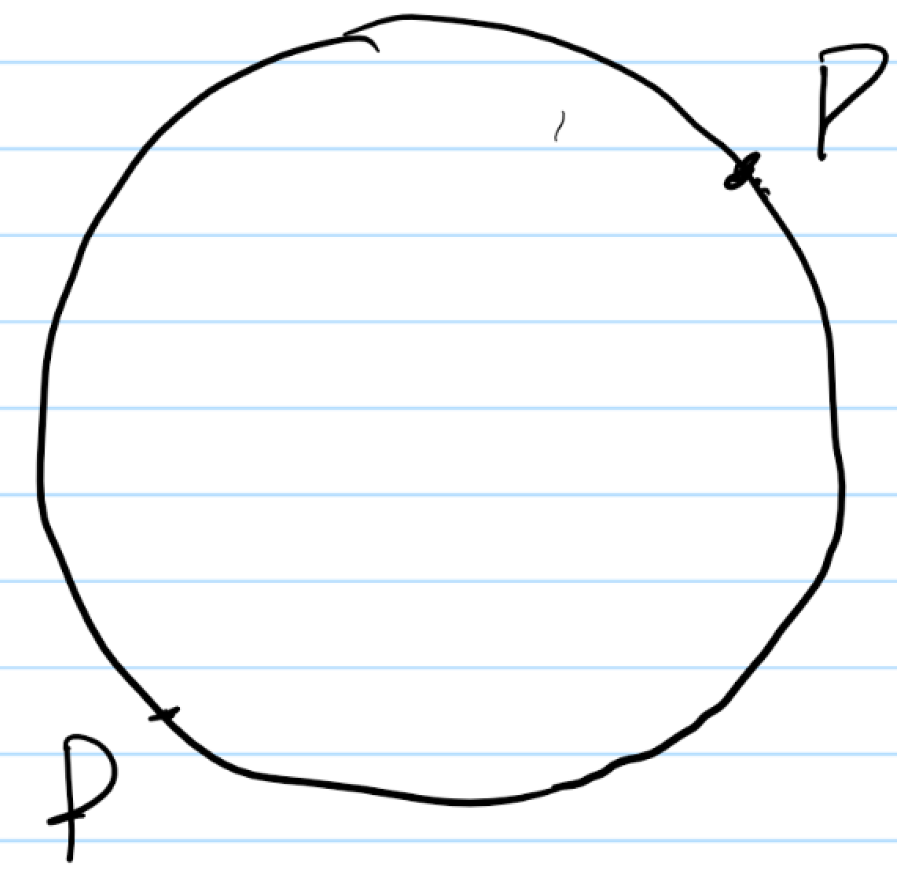
$$\& \ \mathbb{P}^1(\mathbb{R}) = \mathbb{R}^1 \sqcup [(1,0,0)]$$

$$\mathbb{P}^2(\mathbb{R}) = \mathbb{R}^2 \sqcup \mathbb{R}^1 \sqcup \{[(1,0,0)]\}$$

We see $y^2 = x^3 + ax + b$ on $\mathbb{R}^2$ (or $\mathbb{C}^2$)

Rather see $zy^2 = x^3 + axz^2 + bz^3$ in $\mathbb{P}^2(\mathbb{R})$ (or $\mathbb{P}^2(\mathbb{C})$)

This is the projective version of the elliptic curve.



$\mathbb{P}^2(\mathbb{R}) - \mathbb{R}^2$

$[(x,y,z)] \in \mathbb{R}^2$ if $z \neq 0$

$\parallel$

$[(\frac{x}{z}, \frac{y}{z}, 1)]$ if $z \neq 0$

For $z = 0$, $x = 0$, $y = 1$ is a soln

i.e. $[(0,1,0)]$ is a point lying on the Elliptic curve.

This point is called a point at infinity.

$E := \{(x,y) \in \mathbb{R}^2 : y^2 - x^3 - ax - b = 0\} \cup \{\infty\}$

If we see the ~~to see~~ curve in affine space.

$\mathbb{R}^2 \longrightarrow$ space.

If we see the curve in $\mathbb{P}^2(\mathbb{R})$ then

$E = \{[(x,y,z)] \in \mathbb{P}^2(\mathbb{R}) : y^2z - x^3 - axz^2 - bz^3 = 0\}$

Lec 22 Field

$\mathbb{Z}_p$ when p is a prime.

Theorem: a) There is a field of order p^n for any prime p ~~is a prime~~ & ~~$n \in \mathbb{N}$~~ any natural number $n \in \mathbb{N}$

b) ''

LEC 22 Field

$\mathbb{Z}_p$, where p is a prime is a field

Thm: a) There is a field of order p^n for any prime p & any natural number $n \in \mathbb{N}$

b) If $\mathbb{F}$ & $\mathbb{M}$ are fields of order p^n, then $\mathbb{F} \cong \mathbb{M}$

c) If $\mathbb{F}$ is a finite field, then $|\mathbb{F}| = p^n$ for some prime p & $n \in \mathbb{N}$

4

Eg $\mathbb{Z}_p$

$\mathbb{F} \hookrightarrow \left.\dfrac{\mathbb{F}[x]}{\langle f(x)\rangle}\right\}$ is a field

$\mathbb{F}$ is a field $\Rightarrow \mathbb{F}[x]$ is a pid.

$$\frac{\mathbb{R}[x]}{\langle x^2+1\rangle}$$

$$\left(x + \langle x^2+1\rangle\right)^2$$
$$= x^2 + \langle x^2+1\rangle$$
$$= -1 + \langle x^2+1\rangle$$

If $f(x) \in \mathbb{F}[x]$; $\mathbb{F}$ is field is a polynomial of degre

Re. then the quotient $\frac{\mathbb{F}[x]}{}$ is a field as $\langle f(x)\rangle \triangleleft \mathbb{F}[x]$ is